Coding challenge 7: Linear model

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Contents

Question 1: Read the data **1**

Question 2: Fit the full linear model with interaction **3**

the summary of the linear model and ANOVA results: 4

Question 3: simplified linear model without the interaction term **4**

Main results for the simplified model: 5

Question 4: Least-square means for Treatment and Tukey grouping **6**

Interpretation of the Tukey grouping 6

Interpretation 6

Question 5: Plot with letters **7**

What the letters mean 8

Question 6: GitHub **8**

Question 1: Read the data

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
library(multcomp)
```

```
## Loading required package: mvtnorm
## Loading required package: survival
## Loading required package: TH.data
## Loading required package: MASS
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
##
##
## Attaching package: 'TH.data'
##
## The following object is masked from 'package:MASS':
##
##     geyser
```

```r
library(multcompView)

STAND <- read.csv("PlantEmergence.csv")

STAND$Treatment <- as.factor(STAND$Treatment)
STAND$DaysAfterPlanting <- as.factor(STAND$DaysAfterPlanting)
STAND$Rep <- as.factor(STAND$Rep)

str(STAND)
```

```
## 'data.frame':    144 obs. of  7 variables:
##  $ Plot             : int  101 102 103 104 105 106 107 108 109 201 ...
##  $ Treatment        : Factor w/ 9 levels "1","2","3","4",..: 1 2 3 4 5 6 7 8 9 6 ...
##  $ Rep              : Factor w/ 4 levels "1","2","3","4": 1 1 1 1 1 1 1 1 1 2 ...
##  $ Emergence        : num  180.5 54.5 195 198.5 202 ...
```

```
##  $ DatePlanted      : chr  "9-May-22" "9-May-22" "9-May-22" "9-May-22" ...
##  $ DateCounted      : chr  "16-May-22" "16-May-22" "16-May-22" "16-May-22" ...
##  $ DaysAfterPlanting: Factor w/ 4 levels "7","14","21",..: 1 1 1 1 1 1 1 1 1 1 ...
```

Question 2: Fit the full linear model with interaction

```r
lm.full <- lm(Emergence ~ Treatment * DaysAfterPlanting, data = STAND)

summary(lm.full)
```

```
##
## Call:
## lm(formula = Emergence ~ Treatment * DaysAfterPlanting, data = STAND)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -21.250  -6.062  -0.875   6.750  21.875
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     1.823e+02  5.324e+00  34.229   <2e-16 ***
## Treatment2                     -1.365e+02  7.530e+00 -18.128   <2e-16 ***
## Treatment3                      1.112e+01  7.530e+00   1.477    0.142
## Treatment4                      2.500e+00  7.530e+00   0.332    0.741
## Treatment5                      8.750e+00  7.530e+00   1.162    0.248
## Treatment6                      7.000e+00  7.530e+00   0.930    0.355
## Treatment7                     -1.250e-01  7.530e+00  -0.017    0.987
## Treatment8                      9.125e+00  7.530e+00   1.212    0.228
## Treatment9                      2.375e+00  7.530e+00   0.315    0.753
## DaysAfterPlanting14             1.000e+01  7.530e+00   1.328    0.187
## DaysAfterPlanting21             1.062e+01  7.530e+00   1.411    0.161
## DaysAfterPlanting28             1.100e+01  7.530e+00   1.461    0.147
## Treatment2:DaysAfterPlanting14  1.625e+00  1.065e+01   0.153    0.879
## Treatment3:DaysAfterPlanting14 -2.625e+00  1.065e+01  -0.247    0.806
## Treatment4:DaysAfterPlanting14 -6.250e-01  1.065e+01  -0.059    0.953
## Treatment5:DaysAfterPlanting14  2.500e+00  1.065e+01   0.235    0.815
## Treatment6:DaysAfterPlanting14  1.000e+00  1.065e+01   0.094    0.925
## Treatment7:DaysAfterPlanting14 -2.500e+00  1.065e+01  -0.235    0.815
## Treatment8:DaysAfterPlanting14 -2.500e+00  1.065e+01  -0.235    0.815
## Treatment9:DaysAfterPlanting14  6.250e-01  1.065e+01   0.059    0.953
## Treatment2:DaysAfterPlanting21  3.500e+00  1.065e+01   0.329    0.743
## Treatment3:DaysAfterPlanting21 -1.000e+00  1.065e+01  -0.094    0.925
## Treatment4:DaysAfterPlanting21  1.500e+00  1.065e+01   0.141    0.888
## Treatment5:DaysAfterPlanting21  2.875e+00  1.065e+01   0.270    0.788
## Treatment6:DaysAfterPlanting21  4.125e+00  1.065e+01   0.387    0.699
## Treatment7:DaysAfterPlanting21 -2.125e+00  1.065e+01  -0.200    0.842
## Treatment8:DaysAfterPlanting21 -1.500e+00  1.065e+01  -0.141    0.888
## Treatment9:DaysAfterPlanting21 -1.250e+00  1.065e+01  -0.117    0.907
## Treatment2:DaysAfterPlanting28  2.750e+00  1.065e+01   0.258    0.797
## Treatment3:DaysAfterPlanting28 -1.875e+00  1.065e+01  -0.176    0.861
## Treatment4:DaysAfterPlanting28  3.264e-13  1.065e+01   0.000    1.000
```

```
## Treatment5:DaysAfterPlanting28  2.500e+00  1.065e+01   0.235    0.815
## Treatment6:DaysAfterPlanting28  2.125e+00  1.065e+01   0.200    0.842
## Treatment7:DaysAfterPlanting28 -3.625e+00  1.065e+01  -0.340    0.734
## Treatment8:DaysAfterPlanting28 -1.500e+00  1.065e+01  -0.141    0.888
## Treatment9:DaysAfterPlanting28 -8.750e-01  1.065e+01  -0.082    0.935
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.65 on 108 degrees of freedom
## Multiple R-squared:  0.9585, Adjusted R-squared:  0.945
## F-statistic: 71.21 on 35 and 108 DF,  p-value: < 2.2e-16
```

```r
anova(lm.full)
```

```
## Analysis of Variance Table
##
## Response: Emergence
##                              Df Sum Sq Mean Sq  F value    Pr(>F)
## Treatment                     8 279366   34921 307.9516 < 2.2e-16 ***
## DaysAfterPlanting             3   3116    1039   9.1603 1.877e-05 ***
## Treatment:DaysAfterPlanting  24    142       6   0.0522         1
## Residuals                   108  12247     113
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

the summary of the linear model and ANOVA results:

Main results for the full model

ANOVA

Treatment: highly significant, p < 2e-16 DaysAfterPlanting: significant, p = 1.88e-05 Treatment × DaysAfterPlanting: not significant, p = 1.00

What this means

Treatment strongly affects emergence. Days after planting also affects emergence. But the interaction is not significant, so the treatment effect does not appear to change across days after planting.

Question 3: simplified linear model without the interaction term

Do need the interaction term?

No. Since the interaction term is not significant, should remove it and fit the simpler additive model. If the interaction is not significant, simplify the model.

```r
lm.simple <- lm(Emergence ~ Treatment + DaysAfterPlanting, data = STAND)

summary(lm.simple)
```

```
##
## Call:
## lm(formula = Emergence ~ Treatment + DaysAfterPlanting, data = STAND)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.1632  -6.1536  -0.8542   6.1823  21.3958
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          182.163      2.797  65.136  < 2e-16 ***
## Treatment2          -134.531      3.425 -39.277  < 2e-16 ***
## Treatment3             9.750      3.425   2.847  0.00513 **
## Treatment4             2.719      3.425   0.794  0.42876
## Treatment5            10.719      3.425   3.129  0.00216 **
## Treatment6             8.812      3.425   2.573  0.01119 *
## Treatment7            -2.188      3.425  -0.639  0.52416
## Treatment8             7.750      3.425   2.263  0.02529 *
## Treatment9             2.000      3.425   0.584  0.56028
## DaysAfterPlanting14    9.722      2.283   4.258 3.89e-05 ***
## DaysAfterPlanting21   11.306      2.283   4.951 2.21e-06 ***
## DaysAfterPlanting28   10.944      2.283   4.793 4.36e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.688 on 132 degrees of freedom
## Multiple R-squared:  0.958,  Adjusted R-squared:  0.9545
## F-statistic: 273.6 on 11 and 132 DF,  p-value: < 2.2e-16
```

```r
anova(lm.simple)
```

```
## Analysis of Variance Table
##
## Response: Emergence
##                    Df Sum Sq Mean Sq F value    Pr(>F)
## Treatment           8 279366   34921 372.070 < 2.2e-16 ***
## DaysAfterPlanting   3   3116    1039  11.068 1.575e-06 ***
## Residuals         132  12389      94
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Main results for the simplified model:

ANOVA

Treatment: highly significant, p < 2e-16 DaysAfterPlanting: significant, p = 1.58e-06

Important coefficients from the simplified model

Intercept = 182.1632 Treatment2 coefficient = -134.5312

Interpretation of the intercept Because R uses the first level as the reference level, the intercept is the predicted emergence for: Treatment 1 DaysAfterPlanting = 7

So, the expected emergence for Treatment 1 at 7 days after planting is about 182.16 plants.

Interpretation of the coefficient for Treatment 2

The coefficient for Treatment 2 is -134.53. That means, holding DaysAfterPlanting constant, Treatment 2 has about 134.53 fewer emerged plants than Treatment 1, on average.

Question 4: Least-square means for Treatment and Tukey grouping

```r
lsm.trt <- emmeans(lm.simple, ~ Treatment)
lsm.trt
```

```
##  Treatment emmean   SE  df lower.CL upper.CL
##  1          190.2 2.42 132    185.4    194.9
##  2           55.6 2.42 132     50.8     60.4
##  3          199.9 2.42 132    195.1    204.7
##  4          192.9 2.42 132    188.1    197.7
##  5          200.9 2.42 132    196.1    205.7
##  6          199.0 2.42 132    194.2    203.8
##  7          188.0 2.42 132    183.2    192.8
##  8          197.9 2.42 132    193.1    202.7
##  9          192.2 2.42 132    187.4    196.9
##
## Results are averaged over the levels of: DaysAfterPlanting
## Confidence level used: 0.95
```

```r
cld(lsm.trt, alpha = 0.05, Letters = letters, reversed = TRUE)
```

```
##  Treatment emmean   SE  df lower.CL upper.CL .group
##  5          200.9 2.42 132    196.1    205.7  a
##  3          199.9 2.42 132    195.1    204.7  a
##  6          199.0 2.42 132    194.2    203.8  a
##  8          197.9 2.42 132    193.1    202.7  ab
##  4          192.9 2.42 132    188.1    197.7  ab
##  9          192.2 2.42 132    187.4    196.9  ab
##  1          190.2 2.42 132    185.4    194.9  ab
##  7          188.0 2.42 132    183.2    192.8   b
##  2           55.6 2.42 132     50.8     60.4    c
##
## Results are averaged over the levels of: DaysAfterPlanting
## Confidence level used: 0.95
## P value adjustment: tukey method for comparing a family of 9 estimates
## significance level used: alpha = 0.05
## NOTE: If two or more means share the same grouping symbol,
##       then we cannot show them to be different.
##       But we also did not show them to be the same.
```

Interpretation of the Tukey grouping

From the treatment comparisons:

Treatment 2 is much lower than all other treatments. Most of the other treatments are not significantly different from one another. Treatment 5 is significantly higher than Treatment 7. The remaining non-reference treatments mostly overlap statistically.

Interpretation

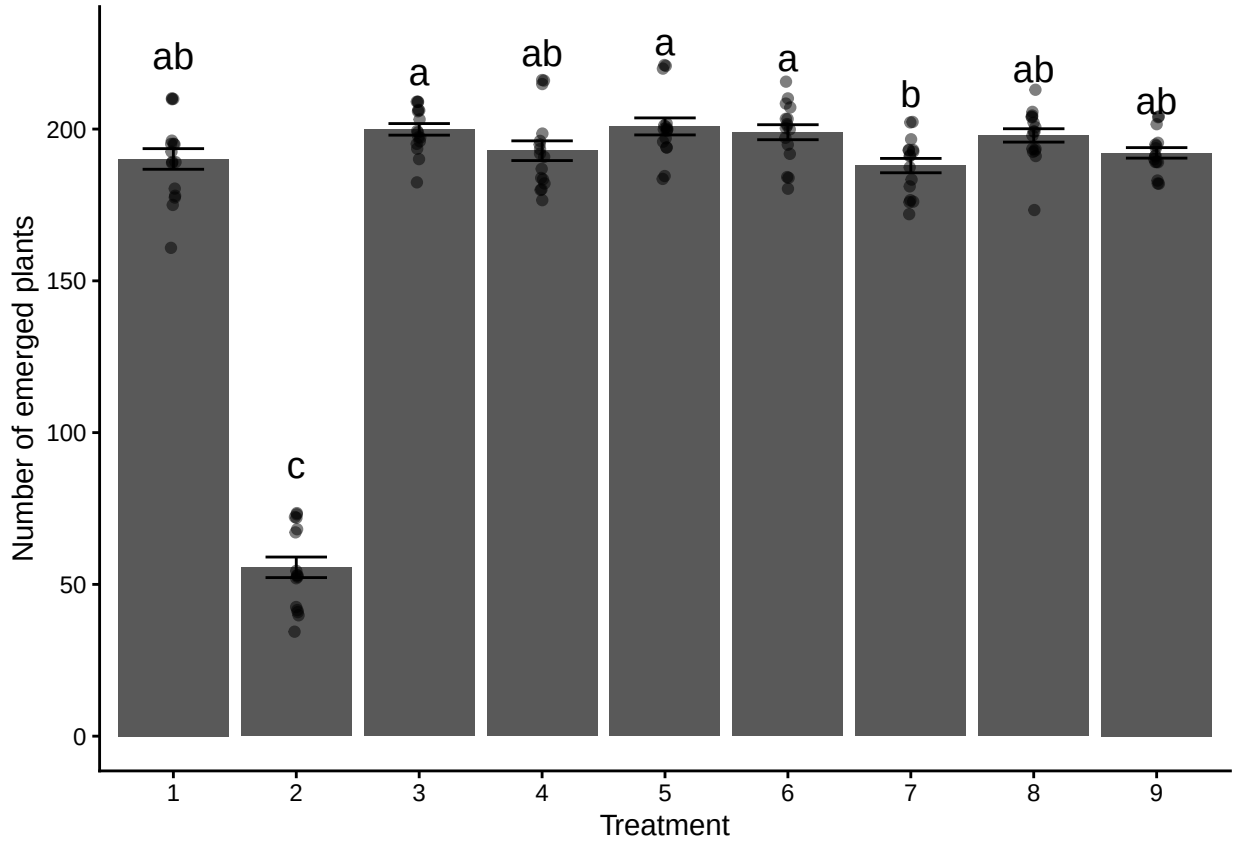
Treatments sharing at least one letter are not significantly different. Treatments with no letters in common are significantly different.

So:

Treatment 2 stands alone as the worst treatment. Treatment 5 is among the highest. Treatment 7 is lower than Treatment 5, but not clearly different from several of the other treatments.

Question 5: Plot with letters

```
plot_cldbars_onefactor <- function(lm_model, factor) {
  data <- lm_model$model
  variables <- colnames(lm_model$model)
  dependent_var <- variables[1]
  independent_var <- variables[2:length(variables)]

  lsmeans <- emmeans(lm_model, as.formula(paste("~", factor)))
  Results_lsmeans <- cld(lsmeans, alpha = 0.05, reversed = TRUE, details = TRUE, Letters = letters)

  sig.diff.letters <- data.frame(Results_lsmeans$emmeans[,1],
                                 str_trim(Results_lsmeans$emmeans[,7]))
  colnames(sig.diff.letters) <- c(factor, "Letters")

  ave_stand2 <- lm_model$model %>%
    group_by(!!sym(factor)) %>%
    dplyr::summarize(
      ave.emerge = mean(.data[[dependent_var]], na.rm = TRUE),
      se = sd(.data[[dependent_var]]) / sqrt(n())
    ) %>%
    left_join(sig.diff.letters, by = factor) %>%
    mutate(letter_position = ave.emerge + 10 * se)

  plot <- ggplot(data, aes(x = !! sym(factor), y = !! sym(dependent_var))) +
    stat_summary(fun = mean, geom = "bar") +
    stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.5) +
    ylab("Number of emerged plants") +
    geom_jitter(width = 0.02, alpha = 0.5) +
    geom_text(data = ave_stand2, aes(label = Letters, y = letter_position), size = 5) +
    xlab(as.character(factor)) +
    theme_classic()

  return(plot)
}
plot_cldbars_onefactor(lm.simple, "Treatment")
```



What the letters mean

The letters show the Tukey groupings:

Bars with the same letter are not significantly different Bars with different letters and no overlap are significantly different

Question 6: GitHub

My GitHub link: